

Design and Loss Analysis of a GaN Half-Bridge Building Block for a Stacked Poly-Phase Bridge

Case Study: EPC2361, 75 V / 5 kW Leg

SPB Building-Block Design Toolkit

September 10, 2026

Abstract

This report documents the electrical, thermal, and layout analysis of a single half-bridge (two-level) building block intended for use as one phase of a stacked poly-phase bridge (modular multicell) inverter, using the EPC2361 enhancement-mode GaN HEMT. Every numeric result quoted here is reproducible from the open-source toolkit accompanying this report (`src/spb.bb/`); this document exists to make the underlying physics, the derivation of every formula used, and the validity of every modeling assumption fully explicit, rather than presenting the results as a black box. Conduction loss is derived in closed form and shown analytically to be independent of modulation index and displacement power factor for synchronously-rectified GaN operation — a result that is proven, not assumed. Switching loss is obtained from a calibrated LTSpice double-pulse test campaign (36 points) run against the vendor’s own SPICE model, not from a datasheet curve fit. The power-loop stray inductance is derived from first principles (Ampère’s law) and checked against three independent published GaN module designs. The recommended design point — four EPC2361 devices in parallel per switch position, $R_g = 1.0\,\Omega$, $f_{sw} \approx 99.3\,\text{kHz}$ — is shown to meet the 99.5% full-load efficiency target with 26% loss-budget margin and 85°C of junction-temperature margin, while two infeasible operating points (1 and 2 parallel devices) are shown, with the same rigor, to be ruled out by real (not assumed) switching-overshoot data.

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1 Introduction and Scope

A stacked poly-phase bridge (a modular multicell / multilevel three-phase inverter) is built from repeated instances of a single *building block*: a two-level half-bridge leg consisting of two switch positions (“top” and “bottom”), each populated by N GaN devices in parallel to reach the required current rating. This report analyzes **one** such building block in isolation, under the following design brief:

- Device: EPC2361 (100 V, 0.75 m Ω typical $R_{DS(on)}$, eGaN HEMT), with the manufacturer’s own LTSpice behavioral model used for switching-loss characterization.
- Modulation: space-vector modulation (SVM) with third-harmonic injection, allowing a modulation index up to the SVM linear-region limit.
- Design targets, supplied by the user and recorded verbatim in `config/design_spec.yaml`, are listed in Table 2.
- Deliverables: (i) the switching frequency that maximizes converter power density while respecting a V_{DS} overshoot limit, (ii) the gate turn-on/turn-off resistance that achieves that frequency, (iii) the number of parallel devices needed, and (iv) proof, not assertion, that the full-load efficiency and thermal targets are met.

Every quantitative claim below is either (a) derived analytically from first principles, with the derivation shown in full, (b) computed by a specific, cited piece of code in the accompanying toolkit, or (c) an LTSpice simulation result against the real vendor device model, with the exact circuit and post-processing method given. Section 10 collects the specific steps taken to *verify* (not merely assume) each of these three categories of claim, including two real bugs that were found and fixed during this work, which are reported here rather than silently corrected, since they materially change how much the reader should trust each intermediate result.

2 Device and Design Specification

2.1 EPC2361 device parameters

All values in Table 1 are taken directly from the EPC2361 datasheet (`data/datasheets/EPC2361_datasheet.pdf`, revision 2.4, 20 July 2026) unless a footnote states otherwise. No datasheet number was estimated, interpolated across an unlabeled axis without saying so, or silently altered.

2.2 Design specification

3 Operating-Point Derivation

3.1 Peak leg current

Real power delivered by a sinusoidal voltage/current pair at displacement power factor $\cos \varphi$ is $P = V_{rms} I_{rms} \cos \varphi$. Solving for the RMS current and converting to peak (crest factor $\sqrt{2}$ for a sinusoid):

$$I_{rms} = \frac{P_{out}}{V_{rms} \cos \varphi}, \quad I_{pk} = \sqrt{2} I_{rms}. \quad (1)$$

Substituting the values from Table 2:

$$I_{rms} = \frac{5000}{30 \times 0.95} = 175.44 \text{ A}, \quad I_{pk} = \sqrt{2} \times 175.44 = 248.1 \text{ A}.$$

This is implemented verbatim in `modulation.leg_peak_current_from_power` and is the single number from which every current-dependent loss and overshoot figure in this report is ultimately derived.

Table 1: EPC2361 parameters used in this analysis.

Parameter	Symbol	Value	Source / test condition
Drain-source voltage rating	$V_{DS,max}$	100 V	absolute maximum
Repetitive transient rating	$V_{DS,tr}$	120 V	duty $\leq 1\%$
Continuous drain current	$I_{D,cont}$	133 A	$T_J < 125^\circ\text{C}$
On resistance (typ)	$R_{DS(on),25}$	0.75 m Ω	$V_{GS}=5\text{V}$, $I_D=50\text{A}$, 25°C
Normalized $R_{DS(on)}$ vs. T_J slope ^a	k_T	$6.4 \times 10^{-3} / ^\circ\text{C}$	fit to Fig. 9, $I_D=50\text{A}$
Gate threshold (typ)	$V_{GS(th)}$	1.1 V	$V_{DS}=V_{GS}$, $I_D=18\text{mA}$
Gate-source charge	Q_{GS}	8.5 nC	$V_{DS}=50\text{V}$, $I_D=50\text{A}$
Gate-drain (Miller) charge	Q_{GD}	3.8 nC	$V_{DS}=50\text{V}$, $I_D=50\text{A}$
Total gate charge (max)	Q_G	34 nC	$V_{DS}=50\text{V}$, $V_{GS}=5\text{V}$
Internal gate resistance	$R_{g,int}$	0.4 Ω	datasheet spec <i>and</i> .lib
Output capacitance	C_{OSS}	1284 pF (max)	$V_{DS}=50\text{V}$, $V_{GS}=0\text{V}$
Energy-equivalent C_{OSS}	$C_{OSS(ER)}$	1419 pF	V_{DS} : 0 $\rightarrow$ 50V sweep
Reverse (3rd-quadrant) voltage	V_{SD}	1.6 V	$I_S=0.5\text{A}$, $V_{GS}=0\text{V}$
Reverse slope resistance ^b	R_{SD}	5 m Ω	fit to Fig. 8
Junction-to-case (top)	$R_{\theta JC}$	0.24 $\Omega \cdot \text{W}^{-1}$ ^c	FEA ladder sum
Junction-to-board	$R_{\theta JB}$	1.5 $\Omega \cdot \text{W}^{-1}$	FEA ladder sum (1.51)
Max junction temperature	$T_{J,max}$	150 $^\circ\text{C}$	absolute maximum

^a Read from Figure 9 (Normalized $R_{DS(on)}$ vs. T_J), which shows the normalized value rising from ≈ 1.0 at 25°C to ≈ 1.8 at 150°C . A linear fit gives $k_T = (1.8 - 1.0)/(150 - 25) = 6.4 \times 10^{-3} / ^\circ\text{C}$, used as

$$R_{DS(on)}(T_J) = R_{DS(on),25} [1 + k_T(T_J - 25)].$$

^b The datasheet's only numeric reverse-conduction point is $V_{SD}=1.6\text{V}$ at $I_S=0.5\text{A}$; the slope R_{SD} is a visual read of Figure 8's 25°C curve ($\approx 2\text{--}5\text{V}$ over $0\text{--}600\text{A}$), i.e. an approximate digitization, not an exact fit. This is flagged explicitly in §11 because it is the least precise input in this report — though also, as shown in §4.6, one of the least consequential.

^c See §7.2 for why 0.24, not the datasheet headline value of 0.2, is used here.

3.2 Full-load loss budget

From the definition of efficiency $\eta = P_{out}/(P_{out} + P_{loss})$, the maximum loss this building block may dissipate at full load while still meeting the efficiency target is

$$P_{loss,budget} = P_{out} \frac{1 - \eta}{\eta}. \quad (2)$$

Numerically: $P_{loss,budget} = 5000 \times \frac{1 - 0.995}{0.995} = 5000 \times 0.005025 = 25.13\text{ W}$, summed over *both* switch positions of the leg (i.e. all $2N$ devices). This is `optimize.loss_budget`, and every “Feasible” verdict in Table 5 is a direct comparison of computed total loss against this single number.

4 Conduction Loss: Closed-Form Derivation

4.1 Topology and the synchronous-rectification assumption

In a hard-switched two-level leg with complementary gating and a short dead time, at any instant *exactly one* of the two switch positions is gated ON (outside the dead-time gap) and carries the full instantaneous leg current $i(\theta)$, $\theta = \omega_1 t$. For a conventional Si MOSFET or IGBT, the ON-state voltage drop differs depending on whether current flows through the channel (forward) or the antiparallel diode (reverse); this asymmetry is the origin of the classical Casanellas result, where switch and diode losses depend on modulation index and power factor separately. GaN e-mode HEMTs have no physical body diode. Reading the EPC2361 LTSpice model directly

Table 2: Design targets, from `config/design_spec.yaml` (user-supplied).

Quantity	Symbol	Value
Real power delivered by this leg	P_{out}	5000 W
DC-link voltage	V_{dc}	75 V
Fundamental RMS output (phase) voltage	V_{rms}	30 V
Displacement power factor	$\cos \varphi$	0.95
Modulation index	M	1.15
Modulation scheme	—	SVM (third-harmonic injection)
Full-load efficiency target	η	99.5%
Dead time	t_{dead}	10 ns
Max. allowed V_{DS} overshoot	—	20% of V_{dc} (15 V)
Ambient temperature	T_a	45°C
T_J design margin below $T_{J,max}$	—	15 °C
TIM + case-to-sink thermal resistance	$R_{\theta,int}$	$0.3 \Omega \cdot W^{-1}$ per device
Heatsink-to-ambient thermal resistance	$R_{\theta,sa}$	$0.4 \Omega \cdot W^{-1}$ (assembly total)
Power-loop geometry (parallel-plate model)	l, w, h	15 mm, 8 mm, 0.2 mm

(`data/ltspace_models/EPC2361.lib`), the channel current is modeled by a single behavioral source

```
bswitch drain source I=if(v(drain,source)>0,
  A1*log(1+exp((v(gate,source)-k2)/k3)) * v(drain,source)/(...),
  -A1*log(1+exp((v(gate,drain)-k2)/k3)) * v(source,drain)/(...))
```

which is explicitly symmetric under ($drain \leftrightarrow source$) exchange: for $v_{DS} < 0$ the *same* functional form re-applies with v_{GD} substituted for v_{GS} . This is the concrete, verifiable basis for the assumption used throughout this section: **a gated-ON EPC2361 conducts through the same resistive channel model regardless of current direction**. Only during the dead-time interval, when *both* gates are off, does the device fall back to genuinely asymmetric (diode-like) third-quadrant conduction, handled separately in §4.5.

4.2 Duty-ratio reference and its validity range

Let the top-switch duty ratio (referenced to the leg midpoint, naturally-sampled modulation) be

$$\text{SPWM: } d_T(\theta) = \frac{1}{2} + \frac{M}{2} \sin \theta, \quad 0 \leq M \leq 1, \quad (3)$$

$$\text{SVM (3rd-harmonic injection): } d_T(\theta) = \frac{1}{2} + \frac{M}{2} \left(\sin \theta + \frac{1}{6} \sin 3\theta \right), \quad 0 \leq M \leq \frac{2}{\sqrt{3}}. \quad (4)$$

Proof that Eq. (4) stays in $[0, 1]$ up to $M = 2/\sqrt{3}$: this is the standard result that adding a 1/6-amplitude third harmonic to a sinusoid reduces its peak magnitude from 1 to $\sqrt{3}/2$, extending the linear (non-saturating) modulation range by a factor $2/\sqrt{3} \approx 1.1547$ — the well-known SVM “hexagon boundary.” The user’s specified $M = 1.15$ is therefore valid (barely: $1.15 < 1.1547$), and the toolkit enforces this bound at run time (`modulation.max_valid_m`, raising a `ValueError` otherwise) rather than silently producing a nonsensical duty ratio outside $[0, 1]$.

The leg current is $i(\theta) = I_{pk} \sin(\theta - \varphi)$, where $\cos \varphi$ is the displacement power factor.

4.3 An orthogonality lemma

Lemma. For integers $n \neq m$ and any real phase c , $\int_0^{2\pi} \sin(n\theta) \cos(m\theta + c) d\theta = 0$.

Proof. By the product-to-sum identity, $\sin(n\theta) \cos(m\theta + c) = \frac{1}{2} [\sin((n+m)\theta + c) + \sin((n-m)\theta - c)]$. Since $n \neq m$, $n-m \neq 0$; and since n, m are treated here as non-negative with $n \neq m$ (and in every application below $n+m > 0$), both $\sin((n+m)\theta + c)$ and $\sin((n-m)\theta - c)$ are sinusoids of nonzero integer frequency in θ , each of which integrates to zero over a full period 2π . The special case $n = 0$ (a constant) against $\cos(m\theta + c)$, $m \neq 0$, integrates to zero directly. ■

4.4 Theorem: synchronous conduction loss is independent of M and φ

Average power dissipated in the top switch position over one fundamental cycle, with a gated-on channel of resistance $R_{DS(on)}$ conducting the full instantaneous current whenever $d_T(\theta)$ indicates it is ON:

$$P_T = \frac{R_{DS(on)}}{2\pi} \int_0^{2\pi} d_T(\theta) i(\theta)^2 d\theta. \quad (5)$$

Expand $i(\theta)^2 = I_{pk}^2 \sin^2(\theta - \varphi) = \frac{I_{pk}^2}{2} [1 - \cos(2\theta - 2\varphi)]$, a sum of a DC term (harmonic 0) and a harmonic-2 term. Using the SVM duty ratio (4) (SPWM is the special case with the $\sin 3\theta$ term dropped), $d_T(\theta)$ is a sum of a DC term (harmonic 0), a harmonic-1 term ($\sin \theta$), and a harmonic-3 term ($\sin 3\theta$). The integrand of Eq. (5) is therefore a sum of $2 \times 3 = 6$ cross terms, pairing $\{0, 1, 3\}$ against $\{0, 2\}$:

	1 (harm. 0)	$-\cos(2\theta - 2\varphi)$ (harm. 2)
$\frac{1}{2}$ (harm. 0)	survives	vanishes (Lemma, $n=0, m=2$)
$\frac{M}{2} \sin \theta$ (harm. 1)	vanishes ($\int \sin \theta d\theta = 0$)	vanishes (Lemma, $n=1, m=2$)
$\frac{M}{12} \sin 3\theta$ (harm. 3)	vanishes ($\int \sin 3\theta d\theta = 0$)	vanishes (Lemma, $n=3, m=2$)

Only the DC×DC term survives, i.e. $\int_0^{2\pi} d_T(\theta) i(\theta)^2 d\theta = \frac{1}{2} \cdot \frac{I_{pk}^2}{2} \cdot 2\pi = \frac{\pi I_{pk}^2}{2}$. Substituting into Eq. (5):

$$P_T = \frac{R_{DS(on)}}{2\pi} \cdot \frac{\pi I_{pk}^2}{2} = \frac{R_{DS(on)} I_{pk}^2}{4} \quad (6)$$

which contains **no dependence on M or φ** , and the identical derivation with the $\sin 3\theta$ term simply absent gives the same result for pure SPWM. By symmetry ($d_B(\theta) = 1 - d_T(\theta)$, and the DC term of $1 - d_T$ is again $1/2$), the bottom switch dissipates the same $P_B = P_T$. **This is a proof, not an approximation:** it follows algebraically from Fourier orthogonality and holds exactly for any M in the valid range and any φ , for a channel that conducts symmetrically when gated on. This is implemented as `modulation.leg_current_stress` and verified by unit test `test_conduction_loss_independent_of_m_and_pf`, which checks the code reproduces this invariance to 9 significant figures for two very different (M, φ) pairs.

Equation (6) can be equivalently written per-device, for N parallel devices sharing current evenly, as

$$P_{T,device} = R_{DS(on)}(T_J) \left(\frac{I_{pk}}{2N} \right)^2. \quad (7)$$

4.5 Dead-time third-quadrant conduction loss

During each dead-time gap (both gates off), the leg current continues to flow, forced through whichever device's third-quadrant path is available, at the higher voltage drop $V_{SD}(I) = V_{SD0} + R_{SD}|I|$ (Table 1). There are two dead-time events per switching period $1/f_{sw}$, so the fraction of the fundamental cycle spent in dead time is $d_{dead} = 2f_{sw}t_{dead}$. The average dead-time loss (summed over the switch position, before splitting by symmetry) is

$$P_{dead} = d_{dead} \cdot \frac{1}{2\pi} \int_0^{2\pi} V_{SD}(|i(\theta)|) |i(\theta)| d\theta. \quad (8)$$

Using the standard result $\frac{1}{2\pi} \int_0^{2\pi} |\sin(\theta - \varphi)| d\theta = \frac{2}{\pi}$ (independent of φ , by periodicity of $|\sin|$), the average rectified current is $\bar{I} = \frac{2}{\pi} I_{pk}$, and

$$P_{dead} = d_{dead} \left(V_{SD0} \bar{I} + R_{SD} \bar{I}^2 \right) \quad (9)$$

where the code evaluates $V_{SD}(\bar{I}) \cdot \bar{I}$ directly (a slightly conservative simplification versus separately averaging $V_{SD0} \bar{I}$ and $R_{SD} \bar{I}^2$, within a few percent given $R_{SD} \bar{I} \ll V_{SD0}$ here). Because $i(\theta) > 0$ for exactly half of every fundamental cycle regardless of φ (a sine wave’s sign pattern is phase-shift invariant in this measure), the dead-time loss also splits exactly 50/50 between the two switch positions, so this term *also* carries no net M/φ dependence at leading order — the asymmetry that exists in a diode-based leg simply does not appear here, because both “sides” of the dead-time conduction are governed by the same symmetric $V_{SD}(I)$ curve.

4.6 Numeric evaluation at the recommended operating point

At the recommended point ($N=4$, $f_{sw}=99.3$ kHz, $T_J \approx 50^\circ\text{C}$):

$$R_{DS(on)}(50^\circ\text{C}) = 0.75 \text{ m}\Omega \times [1 + 6.4 \times 10^{-3}(50 - 25)] = 0.87 \text{ m}\Omega,$$

$$d_{dead} = 2 \times 99.3 \times 10^3 \times 10 \times 10^{-9} = 1.99 \times 10^{-3} \text{ (0.2\% duty)}.$$

Because d_{dead} is only 0.2%, even a $\pm 50\%$ error in the approximately-read R_{SD} (§3, footnote *b*) changes total leg loss by well under 0.1 W — i.e. negligible relative to the 25.13 W budget. This is exactly the kind of explicit uncertainty propagation this report aims to make visible: the least-precisely-known input parameter is also shown, quantitatively, to be the least consequential one. The code (`conduction_loss.py`) reports $P_{cond,leg} = 7.82$ W at this point (Table 5), combining Eq. (7) (both positions, all $2N=8$ devices) with the dead-time term.

5 Switching Loss

5.1 Definition and the two available models

Switching energy is defined, for one turn-on or turn-off transition, as the time integral of instantaneous device power during the transition:

$$E_{on/off} = \int_{t_1}^{t_2} v_{DS}(t) i_D(t) dt, \quad (10)$$

where $[t_1, t_2]$ brackets the transition. Two independent ways of obtaining $E_{on}(I_D, V_{DS}, R_g)$ and $E_{off}(I_D, V_{DS}, R_g)$ are implemented:

1. **Analytical (coarse) model:** scales a single reference point linearly in I_D , V_{DS} , and total gate resistance $R_g + R_{g,int}$ (justified in §5.2). Fast, but does not predict V_{DS} overshoot and is not what produced the final design recommendation in this report.
2. **LTSpice double-pulse lookup model:** the real switching energies from a 36-point sweep against the vendor’s own EPC2361 SPICE model (§5.3), which *is* what produced the final design recommendation.

5.2 Analytical model: justification of each scaling factor

$E_{sw}(I, V, R_g) = E_{ref} \cdot \frac{I}{I_{ref}} \cdot \frac{V}{V_{ref}} \cdot \frac{R_g + R_{g,int}}{R_{g,ref} + R_{g,int}}$, where each factor is justified separately:

- *Linear in I_D* : for a fixed gate resistance, the Miller-plateau clamps V_{GS} and the transition time is set primarily by charging Q_{GD} through R_g , essentially independent of load current to first order (standard gate-charge argument, e.g. Mohan/Undeland/Robbins, *Power Electronics*, Ch.22). Since $E \approx \frac{1}{2} V I t_{transition}$ (triangular V–I overlap) and $t_{transition}$ is roughly I -independent, $E \propto I$.
- *Linear in V_{DS}* : identical argument; $t_{transition}$ is also approximately independent of bus voltage for a MOSFET/HEMT in the gate-charge-limited regime, so $E \propto V$ by the same triangular-overlap approximation.
- *Linear in $R_g + R_{g,int}$* : $t_{transition} \approx (R_g + R_{g,int}) Q_{GD} / \Delta V_{Miller}$ (RC-style charging of the Miller charge), so $t_{transition} \propto R_g$, and therefore $E \propto R_g$.

This is a standard, widely-used first-pass model (the same functional form appears in most vendor loss calculators), explicitly *not* claimed to be accurate for large excursions in R_g or for predicting overshoot — which is precisely why the LTSpice lookup model, not this one, is used for the final recommendation.

5.3 Double-pulse test: circuit topology

The test circuit (generated per-point by `ltspice/double_pulse.py`) is a clamped-inductive double-pulse test using *two instances of the real EPC2361 subcircuit* — the device under test (DUT, low side) and a second instance held permanently off (high side), which provides the free-wheel path via its own third-quadrant conduction, exactly matching how the device will actually operate in this building block (no ideal diode is substituted):

Node	Description
V_{BUS-0}	DC source V_{dc} , with 100 μ F bulk capacitance and 10 m Ω ESR (added for numerical damping, §10)
V_{BUS-A}	L_{loop} , the power-loop stray inductance under test
$A-SW$	high-side device (XQH), held off ($V_{GS}=0$) for the whole test
$A-SW$	L_{load} (100 μ H), the external test inductor setting I_{test}
$SW-SW_{QL-0}$	0 V ammeter, then low-side device (XQL, the DUT), gate driven by the double-pulse waveform through separate $R_{g,on}/R_{g,off}$ paths

The pulse-1 width is sized to reach the target test current from L_{load} 's $V = L dI/dt$ relation: $t_{pw1} = I_{test} L_{load} / V_{dc}$. The second (short) pulse hard-switches that current, and it is around this second pulse's edges that E_{on}/E_{off} are measured.

5.3.1 Why the loop inductance is placed where it is

The fast commutation loop when the DUT switches is $C_{dc} \rightarrow L_{loop} \rightarrow A \rightarrow$ (high-side device) $\rightarrow SW \rightarrow$ DUT $\rightarrow 0 \rightarrow C_{dc}$. L_{load} sits between the *same two nodes* (A, SW) as the high-side device, but because it is $\sim 10^5 \times$ larger than L_{loop} , its current is effectively constant over a single switching transition – it behaves as a current source during commutation, not as part of the fast di/dt loop. L_{loop} , placed between the bulk capacitor and node A , is therefore exactly the parasitic element that determines V_{DS} overshoot, matching standard GaN double-pulse-test practice (full reasoning in `docs/derivations.md`).

5.3.2 Gate drive network

Two ideal SPICE diodes route charge/discharge current through separate resistors:

DRV --[Don]--[Rg_on]--> GL (Don: anode=DRV, conducts when DRV rises)
 GL --[Doff]--[Rg_off]--> DRV (Doff: anode=GL, conducts when DRV falls)

using `.model Dideal D(Ron=1m Roff=1G Vfwd=0)`. **This orientation was initially built backwards** (Doff defined anode-at-DRV instead of anode-at-GL); the symptom, a device that hard-switched on but then never turned off again for the rest of the simulation (zero E_{on}/E_{off} , V_{DS} frozen near 0 V), is documented as Case 1 in §10, together with how it was caught.

5.4 Numerical integration and a second bug: window sizing

E_{on} and E_{off} are each computed by trapezoidal-rule numerical integration (`numpy.trapezoid`) of Eq. (10) over a time window bracketing the relevant edge, read directly off the simulated $V(SW)$ and ammeter-current traces (a dedicated 0 V voltage source is inserted in series with the DUT's drain purely so its branch current gives an unambiguous $i_D(t)$, independent of how the vendor's subcircuit happens to name its internal nodes).

The turn-on window is generous (500 ns, matched to the commanded pulse-2 width) and gave consistent, monotonic results throughout. The *first* implementation of the turn-off window, however, was sized off the ideal gate-drive source's own edge time (2 ns) plus a fixed 5 ns margin – a window of ~ 20 ns *regardless of R_g* . For a slow (large R_g) turn-off, the actual device transition can easily exceed this, and the window silently clipped the tail of the transition. This was caught not by inspection but by a physical-plausibility check on the first full sweep: E_{off} was observed to *collapse* at high R_g (e.g. from $4.05 \mu\text{J}$ at $R_g=1 \Omega$ down to $0.02 \mu\text{J}$ at $R_g=10 \Omega$, for the same I_D) – physically backwards, since a slower turn-off should dissipate *more*, not less, switching energy. The window was widened to $0.6 \times (t_{gap}) \approx 1.2 \mu\text{s}$ (documented as Case 2 in §10), after which E_{off} became monotonically increasing in both R_g and I_D across the full sweep, as physically expected (visible directly in Table 3).

5.5 Full sweep results

The sweep spans $I_D \in \{41.4, 82.7, 124.1, 165.4, 206.8, 248.1\}$ A (the per-device current range for $N \in \{1, \dots, 6\}$ parallel devices at $I_{pk}=248.1$ A) and $R_g \in \{0.5, 1, 2, 4, 7, 10\} \Omega$ (symmetric, $R_{g,on}=R_{g,off}$), at the design's actual $V_{dc}=75$ V and the layout-derived $L_{loop}=0.471$ nH (§6). All 36 points, Table 3, are the complete, unedited output of `scripts/run_epc2361_sweep.py` against the real EPC2361 LTSpice model.

Table 3: Complete LTSpice double-pulse sweep, EPC2361, $V_{dc} = 75$ V, $L_{loop} = 0.471$ nH. E_{on} , E_{off} in μJ ; overshoot in V.

I_D (A)	R_g (Ω)	E_{on} (μJ)	E_{off} (μJ)	V_{DS} overshoot (V)
41.4	0.5	6.93	9.25	56.50
41.4	1.0	8.95	4.05	10.01
41.4	2.0	10.64	5.48	8.64
41.4	4.0	15.50	8.20	6.92
41.4	7.0	20.74	13.71	6.61
41.4	10.0	27.17	19.55	7.04
82.7	0.5	11.05	20.51	76.11
82.7	1.0	10.55	6.24	14.35
82.7	2.0	16.76	10.88	19.90
82.7	4.0	28.19	19.48	14.35
82.7	7.0	40.83	34.87	12.13
82.7	10.0	55.61	50.21	10.06
124.1	0.5	13.79	16.12	42.27
124.1	1.0	18.42	17.79	36.14

I_D (A)	R_g (Ω)	E_{on} (μJ)	E_{off} (μJ)	V_{DS} overshoot (V)
124.1	2.0	28.28	20.50	30.07
124.1	4.0	46.45	34.32	25.53
124.1	7.0	71.59	60.91	14.83
124.1	10.0	95.15	84.59	16.03
165.4	0.5	22.50	24.58	69.17
165.4	1.0	29.16	17.04	54.69
165.4	2.0	42.59	34.84	42.14
165.4	4.0	72.26	52.67	26.34
165.4	7.0	112.69	92.37	20.12
165.4	10.0	148.94	128.11	16.15
206.8	0.5	31.75	35.72	89.80
206.8	1.0	41.32	33.50	69.77
206.8	2.0	62.29	47.48	51.04
206.8	4.0	109.78	76.16	26.88
206.8	7.0	168.09	124.88	24.22
206.8	10.0	211.03	193.73	12.95
248.1	0.5	48.91	36.63	94.05
248.1	1.0	50.70	44.67	75.94
248.1	2.0	88.04	59.53	52.43
248.1	4.0	150.95	98.44	29.05
248.1	7.0	226.90	166.92	22.39
248.1	10.0	297.38	231.61	18.36

Two trends are worth stating explicitly because they are exactly what the underlying physics predicts (a further, independent check on the data’s validity, beyond the monotonicity check that caught the Case-2 bug): (i) for fixed R_g , both E_{on} and E_{off} increase with I_D (more charge moved at higher current); (ii) for fixed I_D , V_{DS} overshoot decreases with R_g (slower di/dt through the same L_{loop}) while E_{on}/E_{off} increase with R_g (slower transitions integrate $v \cdot i$ over more time) – the textbook switching-speed/overshoot trade-off this whole tool exists to navigate.

5.6 From sweep to a usable model: interpolation

Switching loss over the fundamental cycle is obtained by numerically averaging $E_{on}(|i(\theta)|) + E_{off}(|i(\theta)|)$ over $\theta \in [0, 2\pi)$ (720-point discretization) and multiplying by f_{sw} :

$$P_{sw,device} = f_{sw} \left[\frac{1}{2\pi} \int_0^{2\pi} (E_{on} + E_{off})(|i(\theta)|/N) d\theta \right]. \quad (11)$$

$E_{on}(I_D)$, $E_{off}(I_D)$, and $\text{overshoot}(I_D)$ at an arbitrary current not in Table 3 are obtained by interpolating the sweep data. Because the sweep varies only (I_D, R_g) independently (V_{dc} is fixed and $R_{g,on} \equiv R_{g,off}$ throughout), a naïve 4-D Delaunay triangulation over $(I_D, V_{dc}, R_{g,on}, R_{g,off})$ is *geometrically degenerate* (the point cloud is coplanar in 4-D) and SciPy’s Qhull backend fails outright with a precision error – a real failure encountered and fixed during this work (Case 3, §10). The interpolator (`ltspice/lookup.py`) detects this case and reduces to a proper 2-D linear interpolation over (I_D, R_g) , which is both correct for how the data was actually collected and how it is actually queried (`optimize.py` only ever calls the lookup at $R_{g,on} = R_{g,off}$), falling back to nearest-neighbor outside the convex hull of the sweep rather than extrapolating a switching-energy curve linearly (which would be unphysical near the edges of the swept range).

6 Power-Loop Inductance

6.1 Derivation from Ampère’s law

For two parallel conducting planes (or a trace directly over a return plane) of width w , separated by dielectric thickness h , carrying equal and opposite current I over length l , and taking $w \gg h$ (the usual PCB regime, valid here: $w=8\text{ mm} \gg h=0.2\text{ mm}$): approximate the field between the plates as uniform and apply Ampère’s law to a rectangular loop of width w spanning the gap:

$$\oint \vec{H} \cdot d\vec{l} = I_{enc} \Rightarrow H w = I \Rightarrow H = \frac{I}{w}, \quad B = \mu_0 H = \frac{\mu_0 I}{w}.$$

The flux linked by a loop of length l along the current path and height h between the plates is $\Phi = B h l = \mu_0 I h l / w$, so

$$L = \frac{\Phi}{I} = \frac{\mu_0 h l}{w} \quad (12)$$

This is the standard low-inductance “overlapping copper pour” formula used throughout GaN layout literature (e.g. EPC application note AN014 on PCB layout), and is exactly what function `parallel_plate_loop_inductance` in `loop_inductance.py` implements.

6.2 Numeric evaluation

With $h=0.2\text{ mm}$, $l=15\text{ mm}$, $w=8\text{ mm}$ (Table 2) and $\mu_0 = 4\pi \times 10^{-7}\text{ H/m}$:

$$L = \frac{(4\pi \times 10^{-7}) \times (0.2 \times 10^{-3}) \times (15 \times 10^{-3})}{8 \times 10^{-3}} = 4.712 \times 10^{-10}\text{ H} = 0.471\text{ nH},$$

matching the toolkit’s printed output to 3 significant figures.

6.3 Cross-check against published GaN module designs

Table 4 compares this estimate to *measured or field-solved* loop inductances from three independent, peer-reviewed GaN half-bridge module designs (full citations in §13), spanning DBC, insulated-metal-substrate, and PCB constructions.

Table 4: Published GaN half-bridge power-loop inductances vs. this design’s estimate.

Design	Construction	Loop inductance
Passmore et al. (650V/150A) [Ref. 5]	DBC substrate, wire bonds	5.1–5.6 nH
Lu, Chen & Yushyna (650V/120A) [Ref. 5]	Insulated metal substrate	1.3 nH
Lu & Chen (650V/240A) [Ref. 5]	6-layer PCB, flux-canceling	0.7 nH
This design (analytical estimate)	4-layer PCB, parallel-plate model	0.47 nH

The analytical estimate here is *lower* than even the best published, physically measured PCB design (0.7 nH). Equation (12) is exact for the idealized geometry it models (infinite parallel planes, uniform field, $w \gg h$); a real board adds via transitions, connector inductance, and finite-width fringing that this formula does not capture. **This is treated as a known optimism bias in this report, not corrected for numerically** (there is no principled way to add a generic correction factor without a specific layout to field-solve), but it is the single largest source of uncertainty in the recommended R_g and overshoot numbers, and is stated as such in §11 and recommended as the first thing to re-measure once hardware exists.

7 Thermal Model

7.1 Steady-state network

A series thermal-resistance network (Ohm’s-law analogy, $\Delta T = P R_\theta$) is used, appropriate for the full-load steady-state check this report performs (not a transient/ duty-cycled analysis):

$$T_{sink} = T_a + P_{total} R_{\theta,sa}, \quad (13)$$

$$T_{case} = T_{sink} + P_{device} R_{\theta,int}, \quad (14)$$

$$T_J = T_{case} + P_{device} R_{\theta JC}, \quad (15)$$

where $P_{device} = P_{total}/N$ (even current/loss sharing, discussed critically in §11) and P_{total} is the combined conduction+switching loss for one switch position (all N devices). Because $R_{DS(on)}$ depends on T_J (Table 1, footnote *a*) and T_J depends on loss, Eqs. (13)–(15) are solved by fixed-point iteration (3 iterations, `optimize.evaluate_point`), which converges to sub-1°C precision given the mild (0.64%/°C) temperature coefficient.

7.2 Verification: $R_{\theta JC}$ against the FEA Foster ladder

EPC’s companion thermal-model document ([data/datasheets/EPC2361ThermalSPICEModel.pdf](#)) gives a 3-stage Foster RC ladder derived from finite-element simulation, whose resistances sum, by definition, to the steady-state R_θ :

$$R_{\theta JC} = R_{th1} + R_{th2} + R_{th3} = 0.0247 + 0.0421 + 0.173 = 0.2398 \Omega \cdot W^{-1},$$

$$R_{\theta JB} = 0.0683 + 0.693 + 0.749 = 1.5103 \Omega \cdot W^{-1}.$$

The main datasheet table separately lists headline values of 0.2 and 1.5. The $R_{\theta JB}$ ladder sum matches its headline to within 0.7% – consistent with simple rounding. The $R_{\theta JC}$ ladder sum, however, is $\approx 20\%$ *higher* than its headline value – too large to be rounding. **This report uses the more detailed FEA-ladder value, $R_{\theta JC}=0.24 \Omega \cdot W^{-1}$, in preference to the rounded headline figure**, both because it is the more conservative choice and because it is derived from the more detailed underlying simulation. Re-running the full design optimization with this correction (from the originally-used 0.20) changed the recommended point’s T_J by +0.1°C (49.9→50.0°C) – i.e. this discrepancy, once found, turned out not to matter for the final recommendation given the large thermal margin, but it is reported here in the interest of showing the check was actually performed, with its actual (small) consequence, rather than asserting the number is correct without evidence.

7.3 Cooling-path caveat

The datasheet states the package top face is electrically connected to the *source*, and the best thermal path ($R_{\theta JC}=0.24$) is through that top face (EPC’s recommended dual-sided-cooling mount, datasheet Figs. 13–14). In a half-bridge, the high-side and low-side devices’ sources sit at different potentials (switch-node and ground respectively), so a shared top-side heatsink **requires an electrically isolating TIM**. This is not modeled explicitly (the toolkit takes $R_{\theta,int}$ as a user-supplied number), but is flagged here because isolating TIMs typically carry higher thermal resistance than non-isolating ones, and the $R_{\theta,int}=0.3 \Omega \cdot W^{-1}$ used in this report must reflect a real isolating material, not a generic assumption, before this design is trusted on hardware.

8 Optimization Procedure

8.1 Monotonicity lemmas underpinning the bisection searches

Lemma A (overshoot vs. R_g). For fixed I_D , V_{dc} , L_{loop} , V_{DS} overshoot is a monotonically non-increasing function of R_g .

Justification: overshoot arises from $L_{loop} di/dt$ during the turn-off edge; increasing R_g slows the gate charge/discharge rate and hence the achievable di/dt , reducing peak $L_{loop} di/dt$. This is confirmed empirically, not just assumed, by Table 3: for every one of the six I_D rows, overshoot strictly decreases as R_g increases from 1 to 10 Ω (the $R_g=0.5$ point is excluded from this statement – it shows an anomalously large overshoot at several currents, most likely a lightly-damped ~ 200 MHz L_{loop} - C_{OSS} resonance ($f_0 = 1/(2\pi\sqrt{L_{loop}C_{OSS}}) \approx 204$ MHz for $L_{loop}=0.471$ nH, $C_{OSS}\approx 1.3$ nF) being only partially damped at the very fastest edge speed; this point is excluded from design consideration in any case since it violates the overshoot budget by a wide margin at every current tested).

Lemma B (total loss vs. f_{sw}). For fixed R_g and N , total leg loss is strictly increasing in f_{sw} .

Proof: from Eq. (6), conduction loss has no f_{sw} dependence. From Eq. (11), switching loss is f_{sw} times a factor independent of f_{sw} , hence strictly increasing (for nonzero switching energy). Total loss, their sum, is therefore strictly increasing in f_{sw} . ■

These two lemmas are what make the bisection searches in `optimize.py`'s `find_min_rg_for_overshoot` and `find_max_fsw` functions well-posed: each searches a monotonic function for the boundary of a feasibility condition, which is exactly the setting in which bisection is guaranteed to converge to the correct root (given the search interval brackets it).

8.2 Search procedure

For each candidate $N \in \{1, \dots, 6\}$:

1. Compute the per-device peak current I_{pk}/N .
2. Bisect on $R_g \in [0, 10] \Omega$ (Lemma A) for the smallest R_g such that interpolated overshoot $\leq 20\% V_{dc} = 15$ V. If even $R_g=10 \Omega$ exceeds the limit, N is infeasible and is excluded from the results table (this is exactly what happens for $N=1, 2$; see §9).
3. At that R_g , bisect on $f_{sw} \in [10, 100]$ kHz (the user's configured search range; Lemma B) for the largest f_{sw} such that total loss $\leq$ the 25.13 W budget and $T_J \leq T_{J,max} - 15^\circ\text{C} = 135^\circ\text{C}$ (Eqs. (13)–(15)).

9 Results

Table 5: f_{sw} vs. parallel-device-count trade-off (feasible points only).

N	R_g (Ω)	f_{sw} (kHz)	P_{cond} (W)	P_{sw} (W)	P_{total} (W)	Budget (W)	T_J ($^\circ\text{C}$)	Margin ($^\circ\text{C}$)	V_{DS} ovs. (V)
3	3.8	76.1	9.93	15.07	25.00	25.13	52.3	82.7	14.8
4	1.0	99.3	7.82	10.88	18.70	25.13	50.0	85.0	12.2
5	1.0	99.3	6.46	13.11	19.57	25.13	50.0	85.0	10.9
6	0.8	99.3	5.55	15.49	21.05	25.13	50.2	84.8	10.0

$N = 1$ and $N = 2$ are absent from Table 5 because they are infeasible, not because they were not searched: at $I_{pk}=248.1$ A ($N=1$) and 124.1 A ($N=2$), Table 3 shows overshoot exceeds the 15 V limit even at the maximum searched $R_g = 10 \Omega$ (18.4 V and 16.0 V respectively). This is a conclusion drawn directly from real simulated switching data, not from a rule of thumb.

9.1 The f_{sw} ceiling at $N \geq 4$

$N=4, 5, 6$ all converge to $f_{sw} \approx 99.3$ kHz, i.e. essentially the search's *upper bound* (`fsw_max_Hz` = 100 kHz in `config/design_spec.yaml`), not a value found in the interior of the search interval. This is visible directly in the loss-budget utilization column: $N=4$ uses only 74% of the loss budget and retains 85°C of thermal margin at that frequency – i.e. the bisection hit the *user-configured ceiling*, not a physical loss or thermal limit. If higher switching frequency (and correspondingly smaller magnetics) is desired, `fsw_max_Hz` should be raised and the optimization re-run; this report does not claim to know the true physical maximum for $N \geq 4$ because it was never searched.

9.2 Recommended design point

$N=4$ is recommended over $N=5, 6$ because it achieves the *same* search-bound-limited frequency with fewer devices (lower BOM cost and board area) and a healthy 26% loss-budget margin, and over $N=3$ because $N=3$ is genuinely loss-limited (99.7% budget utilization, overshoot at 14.8 V against a 15 V limit – effectively no design margin) and achieves a much lower f_{sw} (76.1 kHz) as a direct consequence:

$$\mathbf{N=4}, \quad R_g = 1.0 \, \Omega \text{ (on and off)}, \quad f_{sw} \approx 99.3 \text{ kHz},$$

$$P_{loss} = 18.70 \text{ W (74\% of budget)}, \quad T_J = 50.0^\circ\text{C (85}^\circ\text{C margin)}, \quad V_{DS} \text{ overshoot} = 12.2 \text{ V}.$$

10 Model Verification: Every Check That Was Actually Performed

This section exists because a formula being derived correctly does not guarantee it was *implemented* correctly, or that its inputs (a vendor SPICE model, a datasheet figure) were read correctly. Every discrepancy below was found by an explicit check, not assumed away.

Case 1 – Gate-drive diode orientation (double-pulse netlist). The turn-off discharge diode was initially wired anode-at-DRV instead of anode-at-GL, meaning the gate could charge but never discharge. *Symptom*: a smoke test against a generic placeholder MOSFET model returned $E_{on} = E_{off} = 0$ and $V(SW)$ frozen near 0 V for the entire simulation – physically impossible for a circuit that is supposed to be switching. *Fix*: reversed the diode (§5.3). *Re-verification*: re-ran the same smoke test; $V(SW)$ correctly settled to $V_{dc} + V_f$ during the freewheel window and non-zero, current-dependent E_{on}/E_{off} were obtained.

Case 2 – Turn-off integration window (§5.4). Caught by the same kind of physical-plausibility check applied to the full sweep: E_{off} decreasing with R_g is backwards. Fixed by widening the window; re-verified by confirming E_{off} became monotonically increasing in R_g across all six currents in the corrected sweep (Table 3).

Case 3 – Degenerate 4-D interpolation. The first attempt to build a switching-loss lookup model from the sweep raised a `QhullError` (“Initial simplex is flat”). *Root cause*: the sweep varies only 2 of the 4 nominal input dimensions independently (V_{dc} fixed, $R_{g,on} \equiv R_{g,off}$), making the 4-D point cloud coplanar. *Fix*: detect this condition and reduce to a well-posed 2-D interpolation (§5.5).

Case 4 – Pin order. The EPC2361 subcircuit is instantiated as `XQ {drain} {gate} {source}` EPC2361 or similar in a double-pulse netlist; getting this wrong produces a netlist that *simulates without error* but models a nonsensical circuit. Verified independently two ways: (i) the `.subckt EPC2361 gatein drainin sourcein` declaration in `EPCGaNLibrary.lib` itself, and (ii)

the PINATTR `SpiceOrder` entries in the companion `EPCGaN.asy` symbol file (1=gatein, 2=drainin, 3=sourcein). Both agree: gate–drain–source, not drain–gate–source as this project’s own default initially assumed.

Case 5 – $R_{\theta JC}$ discrepancy. Described in full in §7.2: found by summing the FEA Foster ladder and comparing to the datasheet headline value, revealing a real (not rounding) $\approx 20\%$ gap; resolved by using the more detailed, more conservative value.

Case 6 – LTSpice numerical convergence. The very first full sweep attempt intermittently produced “Heightened Def Con” Newton-iteration warnings and, on one point ($R_g=0.5\ \Omega$, the fastest/highest- di/dt case), timed out after 600s without converging. This is consistent with a lightly-damped ~ 200 MHz L_{loop} – C_{OSS} resonance (§8) stressing the behavioral (exponential/logarithmic) device model’s Newton solver. *Fix*: relaxed solver tolerances (`reltol=0.01`, up from LTSpice’s default `1e-3`, a standard mitigation documented in GaN-vendor LTSpice application notes for exactly this class of stiff behavioral-model problem) and added a small (10 m Ω) physically-realistic ESR in series with the bulk capacitor for numerical damping. *Re-verification*: the full 36-point sweep then completed without a single failed or non-converged point.

Unit test coverage. Six automated tests (`tests/test_loss_models.py`) directly assert several of the theorems and monotonicity claims made in this report: (i) conduction loss independence from M and φ to 9 significant figures (§4.4); (ii) conduction loss halving under exact $2\times$ device paralleling for the resistive term; (iii) switching loss doubling under exact $2\times f_{sw}$ (Lemma B); (iv) switching loss increasing with R_g in the analytical model; (v) loop-inductance formulas returning positive, order-of-magnitude-sane values; (vi) the Casanellas cross-check (§5.2 discussion) reconstructing the synchronous-model total RMS current to within 30%. These tests were run, and passed, after every code change described in this report, not only once at the end.

11 Assumptions: Summary and Confidence Assessment

Table 6: Every modeling assumption used in this report, with its justification and an honest confidence assessment.

Assumption	Justification / evidence	Confidence
Synchronous, symmetric-channel GaN conduction (no diode asymmetry when gated on)	Directly verified against the EPC2361 <code>.lib</code> behavioral source, which is symmetric under drain/source exchange (§4)	Proven (for this device model)
Conduction loss independent of M, φ	Closed-form proof via Fourier orthogonality (§4.4), unit-tested to 9 s.f.	Proven
Switching loss data (Table 3)	Real LTSpice simulation of the vendor’s own device model, not a datasheet curve fit; internally consistent monotonic trends in both I_D and R_g	High (simulation, not hardware)
Steady-state, full-load-only thermal check	No transient/duty-cycled T_J model implemented	Stated limitation , not a hidden gap
Loop inductance, 0.47 nH	Derived from Ampère’s law (exact for the idealized geometry); <i>lower</i> than the best of 3 published, physically-measured PCB designs (0.7–5.6 nH)	Optimistic – explicitly flagged, not corrected

Assumption	Justification / evidence	Confidence
Even current sharing across N parallel devices	<i>Not</i> verified; a cited experimental paper (Palma et al., PCIM 2024) shows real boards see dynamic imbalance from V_{th} spread	Acknowledged simplification , real designs need layout symmetry this model cannot check
R_{SD} (3rd-quadrant slope resistance)	Approximate visual read of an undigitized datasheet curve	Low precision, but shown negligible to the result (§4.6: < 0.1 W impact)
$R_{\theta JC} = 0.24$, not the datasheet headline 0.2	FEA Foster-ladder sum; found by explicit cross-check, changes recommended T_J by $+0.1^\circ\text{C}$	Verified and corrected
Isolating TIM assumed for $R_{\theta, int}$	Required by device package construction (top = source potential); not independently confirmed	User must verify against their actual TIM choice

12 Limitations and Recommended Next Steps

1. Re-run the double-pulse sweep at a more conservative loop inductance (1–2 nH, per §6.3) before committing to $R_g = 1.0\ \Omega$ for a first hardware revision.
2. If a smaller magnetics / higher power density is wanted, raise `fsw_max_Hz` beyond 100 kHz and re-run: §9 shows $N=4$ –6 have substantial unused loss and thermal margin at the current ceiling.
3. Confirm the actual TIM part number’s thermal resistance and electrical isolation rating against §7.2’s cooling-path discussion.
4. Once hardware exists, replace the analytical loop-inductance estimate with a measured value (impedance analyzer or a ringing-based extraction, as used by two of the three benchmark papers in Table 4) and re-run the sweep.
5. Extend the thermal model from steady-state to the FEA-derived 3-stage Foster ladder (already recorded in `data/devices/EPC2361.yaml` but not yet consumed by `thermal.py`) if duty-cycled or transient operation needs to be checked.

13 References

1. EPC2361 datasheet, rev. 2.4, Efficient Power Conversion Corp., 20 July 2026.
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7. N. Mohan, T.M. Undeland, W.P. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd ed., Wiley, 2003 (gate-charge-limited switching-time argument, §5.2).